Hyperbolic Polytope Volume as an LLM Benchmark: Constructive Geometry Meets Numerical Verification

Igor Rivin

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Abstract

We propose a family of benchmark problems for evaluating the mathematical reasoning capabilities of large language models (LLMs): given N points on S^{d-1}, construct an ideal polytope in hyperbolic d-space $\mathbb{H}^d$ with a specified geometric measure (volume, surface area, or isoperimetric ratio) within a prescribed tolerance window. Unlike traditional benchmarks that test symbolic manipulation or theorem proving, these problems require the model to *construct* a geometric object and produce a verifiable numerical certificate. We evaluate three frontier models (Claude, GPT-5.4 Pro, and Gemini 3.1 Pro DeepThink) on problems in $\mathbb{H}^4$ and $\mathbb{H}^6$, revealing qualitatively different failure modes: GPT-5.4 Pro excels at persistent numerical construction but occasionally misinterprets the problem; Gemini produces elegant algebraic reasoning but fabricates numerical results; Claude's toolkit provides brute-force optimization with certified answers. A standalone checker based on the Haagerup–Munkholm formula ($d = 4$) and the Hopf reduction with Murakami dilogarithms ($d = 6$) enables fully automated verification. As a byproduct of the evaluation, we discover a combinatorial invariance phenomenon: the f-vector of the volume-maximizing polytope is the same whether volume is measured in the Euclidean or hyperbolic sense, across dimensions $d = 3, 4, 5, 6$.

1 Introduction

The rapid development of large language models has created a need for benchmarks that test genuine mathematical reasoning, not just pattern matching or memorization. An ideal benchmark should:

(i) have a *verifiable* answer that can be checked automatically;

(ii) require *construction*, not just classification or selection;

(iii) be *parameterizable*, so that memorization of specific instances does not help;

(iv) involve *non-trivial mathematics* that is unlikely to appear verbatim in training data.

We propose that the following family of problems satisfies all four criteria:

Problem. *Given integers N and d, and a target value T with tolerance ϵ, produce N points $v_1, \ldots, v_N \in S^{d-1} \subset \mathbb{R}^d$ such that the volume of the ideal polytope* $\mathrm{conv}(v_1, \ldots, v_N) \subset \mathbb{H}^d$ *lies in the interval $[T - \epsilon, T + \epsilon]$.*

Variants include prescribing the number of facets (rather than vertices), optimizing boundary surface area or the isoperimetric ratio V/A, and working in the Euclidean rather than hyperbolic setting.

The key feature is that the answer is a *numerical certificate*: a list of coordinates that can be independently verified by a standalone checker implementing the relevant volume formula. No human judgment is required to grade the response.

2 The Verification Pipeline

2.1 Volume formulas

For $d = 4$, the volume of an ideal 4-simplex $[\infty, v_1, v_2, v_3, v_4]$ in the upper half-space model is given by the Haagerup–Munkholm formula:

$$\mathrm{Vol}_4 = \frac{\pi}{3}\left(4\pi - \sum_{i<j}\varphi_{ij} - \sum_{\ell}\arccos\frac{d_\ell}{R}\right),$$

where φ_{ij} are the Euclidean dihedral angles of the tetrahedron $T = \mathrm{conv}(v_1, \ldots, v_4)$, R is its circumradius, and d_ℓ are signed distances from the circumcenter to the faces.

For $d = 6$, ideal simplices have vanishing vertex links ($w_0 = 0$), and the Hopf reduction gives:

$$\mathrm{Vol}_6(\Delta_{\mathrm{ideal}}) = \frac{8\pi^3}{15}\left(\frac{17}{4} - w_4 + \frac{1}{2}w_2\right),$$

where $w_4 = \frac{1}{2\pi}\sum_{i<j}\theta_{ij}$ involves the 21 dihedral angles, and $w_2 = \frac{1}{2\pi^2}\sum_{|I|=4}\mathrm{Vol}_{S^3}(L_I)$ involves the 35 spherical tetrahedral link volumes, each computable by Murakami's dilogarithm formula.

2.2 The standalone checker

The checker (`standalone_checker.py`) is a self-contained Python script (no external dependencies beyond `numpy` and `scipy`) that verifies:

1. All N vertices lie on S^{d-1} (tolerance 10^{-6}).
2. All vertices are on the convex hull (no interior points).
3. The convex hull has the prescribed number of facets (if specified).
4. The computed geometric measure falls within $[T - \epsilon, T + \epsilon]$.

Exit code 0 indicates PASS; exit code 1 indicates FAIL. The checker supports seven measure types: `volume` ($\mathbb{H}^4$), `surface_area` ($\mathbb{H}^4$), `ratio` ($\mathbb{H}^4$), `volume_H6`, `euclidean_volume`, `euclidean_area`, and `euclidean_ratio`.

3 Model Evaluation

We evaluated three frontier models on a series of geometric construction tasks in $\mathbb{H}^4$ and $\mathbb{H}^6$. Each model was given the same prompt asking for the best lower bound on the maximal volume, with the expectation that the model would produce explicit vertex coordinates.

3.1 Summary of results

3.2 Qualitative comparison

The three models exhibit qualitatively different strengths and failure modes.

Table 1: Model performance on selected benchmark tasks. Bold indicates the best answer. A dash indicates no numerical answer was produced.

Task	Best known	Claude	GPT-5.4 Pro	Gemini 3.1
$\mathbb{H}^4$ vol, $N = 30$	22.262	**22.262**	22.107	—
$\mathbb{H}^4$ vol, 20 simpl. facets	2.3139	**2.3139**	2.3139	1.61*
Eucl. vol, $N = 30$ on S^3	2.4909	**2.4909**	2.4839	—
Eucl. vol, 20 facets on S^3	0.7925	**0.7925**	**0.7925**	**0.7925**†
Eucl. area, 20 facets on S^3	5.9544	**5.9544**	**5.9544**	**5.9544**†
Eucl. area, 17 facets on S^3	5.036	5.002	**5.036**	—
$\mathbb{H}^6$ vol, $N = 16$	0.5406	**0.5406**	pending	0.041*

*Gemini's answer is formally correct but useless (off by a factor of 13 for $\mathbb{H}^6$; stacked-simplex bound for 20 facets).
†Gemini provided the correct algebraic formula but no vertex coordinates, and falsely claimed numerical optimization confirmation.

GPT-5.4 Pro: persistent numerical constructor. Given a problem, GPT-5.4 Pro typically produces explicit coordinates, verifies them by multiple methods (e.g., star triangulations from all vertices), and states honest caveats ("I have not proved this is optimal"). It identified the free sum $C_5 \oplus I \oplus I$ as the Euclidean volume maximizer for 20 simplicial facets and derived the exact formula $V = \frac{5}{24}\sqrt{10 + 2\sqrt{5}}$. Its main weakness is occasional misinterpretation of the problem (e.g., reading $\mathbb{H}^6$ as the 6-cube in one instance, despite correctly interpreting $\mathbb{H}^4$ in a dozen previous conversations), which is easily corrected by re-prompting. On the 17-facet surface area problem, GPT-5.4 Pro found a configuration (area 5.036) that our 1000-restart optimizer could not beat.

Gemini 3.1 Pro DeepThink: strong algebraic reasoner, unreliable numerical claims. Gemini independently derived the exact formula $A = \frac{5}{3}\sqrt{15 - \sqrt{5}}$ for the 20-facet surface area, and proposed creative constructions (the 10-10 duoprism on the Clifford torus for 20 non-simplicial facets). However, it claimed that "random-initialization geometric optimization confirms" a numerical value of 12.4417 for the $N = 30$ surface area; when asked to produce the optimization code, it spent over an hour and delivered code that only reaches 12.423. When confronted about the discrepancy, it entered a computation that ran for over 16 hours with no output. Its $\mathbb{H}^6$ answer ($4v_6 < M < 220v_6$) is a correct but useless interval spanning a factor of 55. On a separate occasion, Gemini produced a long, beautifully structured essay that misunderstood the optimization direction entirely (minimizing the supremum instead of maximizing the volume).

Claude (with toolkit): brute-force optimizer with certified answers. Using multi-start L-BFGS-B with 500+ restarts and a standalone verification pipeline, Claude's toolkit provides the strongest numerical results on most tasks but lacks the algebraic insight that GPT-5.4 Pro and Gemini occasionally provide. The hybrid approach—using Euclidean warm starts followed by hyperbolic polishing—was discovered during this evaluation and reduces optimization time by 38%. Several code improvements were adopted from the other models' suggestions: the Cartesian parameterization with norm projection (from Gemini 3.1's basin-hopping code) and the Powell optimizer for facet-constrained problems (from GPT-5.4 Pro's code).

3.3 The "windowed" benchmark format

A key design choice is the *windowed* format: the target value T is chosen from the 70th–95th percentile of the optimization landscape, not the global maximum. This prevents the model from simply maximizing the objective (which would give the same answer for every instance) and instead requires it to *control* the geometry to hit a specific window. This is substantially harder: a model that can only maximize cannot pass a windowed test with a target below the maximum.

We generated 66 windowed benchmark cases (33 hyperbolic, 33 Euclidean) covering volumes, surface areas, and isoperimetric ratios for $N = 10, 20, 30$ vertices, including facet-count-constrained variants.

3.4 Lessons for LLM training

The evaluation suggests that current training procedures produce characteristically different failure modes:

- **GPT-5.4 Pro** appears trained to minimize type II errors (failure to produce useful output), resulting in persistent numerical construction with honest uncertainty quantification.
- **Gemini** appears trained to minimize type I errors (false specific claims), resulting in a tendency to produce formulas without evaluating them, and to fabricate numerical confirmation when pressured for specific answers.
- The ability to produce *verifiable certificates* (code plus vertex coordinates) separates genuine computation from confabulation. Our checker enforces this distinction automatically.

4 Mathematical Byproducts

The LLM evaluation process, combined with systematic computational exploration, led to several mathematical discoveries.

4.1 Combinatorial invariance

The f-vector of the volume-maximizing ideal polytope is the same whether volume is measured in the Euclidean or hyperbolic sense. This was verified in $d = 3$ ($N \leq 50$), $d = 4$ ($N \leq 30$), and $d = 6$ ($N \leq 24$), with no exceptions. The invariance holds both within the regime where optimal polytopes are product polytopes and beyond it, suggesting a universal phenomenon.

4.2 Identification of optimal polytopes

For small N in even dimension $d = 2k$, the volume-optimal polytope is the join of k regular polygons in orthogonal 2-planes: $C_{p_1} \oplus \cdots \oplus C_{p_k}$. This is optimal up to $N \approx 2d + 2$, beyond which asymmetric configurations take over. Notable cases include the cross-polytope ($N = 2d$) and the simplex ($N = d + 1$). Volume-optimal polytopes are not cyclic polytopes and are in fact far from neighborly: at $N = 30$ in $d = 4$, the optimal polytope has only 123 facets compared to 756 for the cyclic polytope $C(30, 4)$.

4.3 The volume plateau

At $N = 9$ in $d = 4$ (equivalently, 20 simplicial facets), the hyperbolic volume maximum is achieved on a manifold of dimension ≥ 2, not an isolated point. The Euclidean maximizer (the free sum $C_5 \oplus I \oplus I$, with exact volume $\frac{5}{24}\sqrt{10 + 2\sqrt{5}}$) and the hyperbolic maximizer are metrically distinct configurations with identical hyperbolic volume to 13 significant digits. For prime facet counts (e.g., 17), the maximum is isolated.

4.4 Volume–volume correlation

Across the optimization landscape, the Euclidean and hyperbolic volumes are correlated at $r > 0.995$, with log-log elasticity approaching 2 at $N = 30$ in $d = 4$. This near-proportionality is surprising given that the Euclidean volume is polynomial in the vertex coordinates while the hyperbolic volume involves the transcendental Haagerup–Munkholm formula. For random configurations (as opposed to near-optimal ones), the correlation is lower ($r \approx 0.86$ in $d = 6$) but still substantial, and holds even within a fixed combinatorial type ($r \approx 0.88$ within a single facet-count bucket).

4.5 Distribution of random polytope volumes

For N points sampled uniformly on S^{d-1}, both the Euclidean hull volume and the ideal $\mathbb{H}^d$ polytope volume are random variables whose distribution depends on N and d. In $d = 6$, we find a transition in the distribution shape as N grows:

N	Skewness	Normal p	Beta p	Best fit
10	$+0.78$	< 0.001	0.985	Beta$(3.9, 43.7)$
12	$+0.65$	0.005	0.766	Beta$(6.9, 120.7)$
16	$+0.23$	0.088	< 0.001	Borderline
20	$+0.02$	**0.832**	< 0.001	Normal
30	-0.09	0.281	< 0.001	Normal
50	-0.19	0.034	< 0.001	Normal (slight left skew)

For small N (≤ 12), the volume distribution is well-described by a Beta distribution with parameters scaling linearly in N, consistent with earlier observations in $d = 3$ and $d = 4$. For large N (≥ 20), the distribution converges to Normal—as expected from a CLT-type argument, since the polytope volume becomes a sum of many weakly-dependent simplex volumes in the Delaunay triangulation. The crossover occurs at $N \approx 16$–20 in $d = 6$ (where the skewness crosses zero), and is accompanied by a subtle transition from right skew to slight left skew.

The within-bucket distributions (conditioning on a fixed facet count) show essentially the same shape and correlation structure, confirming that the distributional properties are genuinely geometric rather than combinatorial artifacts.

5 Conclusions

The geometric construction benchmark exposes qualitative differences between frontier LLMs that traditional math benchmarks miss. The ability to produce *verifiable certificates*—explicit coordinates that pass an independent numerical checker—is a fundamentally different capability from producing plausible-sounding mathematical text.

The benchmark is parameterizable (any N, d, target, tolerance), resistant to memorization (windowed targets from a continuous landscape), and fully automatically gradable (standalone checker with exit codes). The mathematical content—hyperbolic volume, Murakami dilogarithms, Hopf reduction—is sufficiently specialized that training-data contamination is unlikely.

As a byproduct, the evaluation process itself generated new mathematical results (combinatorial invariance, identification of optimal polytopes, volume plateau), demonstrating that the "mixture of experts" approach —combining human mathematical intuition with LLM algebraic insight and automated numerical optimization—can be more productive than any single participant acting alone.

References

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